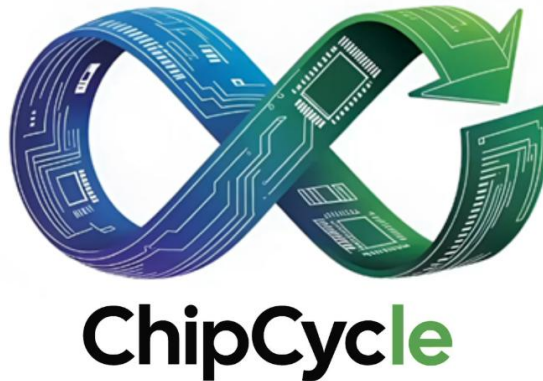


ENGLISH, ETHICS & EMPLOYABILITY

FINAL INTEGRATIVE PROJECT



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SECTION 1 . CHALLENGE STATEMENT

The Semiconductor Circularity Paradox:

An Ethical, Regulatory, and Strategic Imperative for Global Resilience

1.1 THE PROBLEM: A Convergent Crisis with National Security Implications

The semiconductor industry operates on a 19th-century linear resource model that directly contradicts its 21st-century technological ambitions and U.S. strategic objectives under the CHIPS and Science Act.

- **Geopolitical Material Vulnerability = National Security Risk:**

Supply chains are strategically compromised. China controls 98% of primary gallium production and 90–95% of rare earth refining (USGS, 2024). These dependencies are now active instruments of geopolitical leverage. For the U.S., achieving CHIPS Act objectives is impossible without securing alternative, resilient, circular, and domestically accessible material loops.

- **Industrial-Scale Environmental Impact = Compliance & Reputation Risk:**

Manufacturing resource intensity is unsustainable: advanced 300mm (about 11.81 in) wafers require 3,800–4,200 liters (about 1109.52 gal) of ultrapure water per unit (Benner/NCCAVS, 2024). Primary extraction generates 2,000 tons of toxic tailings per ton of rare earths (Kaya, 2021), creating liabilities that violate ESG mandates and community trust.

- **The Manufacturing Intelligence Deficit = Competitive Risk:**

Each scrapped chip represents lost materials and lost process intelligence. The industry forfeits tens of billions annually as the vast majority of scrap enters opaque “blind markets” with limited traceability. Industry analyses (SEMI, IMEC, 2024) argue this data black hole prevents the chain of custody, root cause analysis, and continuous improvement required for technological leadership and regulatory compliance

1.2 WHY IT MATTERS: The Strategic Inflection Point (2024–2028)

Circularity has evolved from a CSR initiative to a mandatory condition for market access and competitive survival.

DRIVER	IMPACT	CONSEQUENCE OF INACTION
U.S. CHIPS Act	Funding tied to sustainable supply chains	Loss of federal subsidies & erosion of domestic advantage
EU Product Passport	Progressive deployment of product-level traceability requirements from 2027	Risk of exclusion from a €500B+ electronics market in the EU if requirements are not met
California SB-253	Mandatory Scope 3 emissions disclosure	Heightened reputational and investor risk
Corporate Procurement	Circularity mandates from Apple, Google, etc.	Loss of major Tier-1 customer accounts

Table 1 Regulatory and Commercial Drivers

1.3 MULTI- STAKEHOLDER IMPACT ANALYSIS

"The semiconductor industry cannot achieve CHIPS Act ambitions with a linear model. Circular intelligence is the strategic prerequisite for technological sovereignty."

- **Semiconductor Fabs:** Production can reach (\$5-10M/day) and yield losses. **Key focus:** CHIPS Act compliance - Securing domestic resilience meeting CHIPS-related conditions.
- **Electronics Brands:** Supply chain instability and greenwashing risks. **Key focus:** Meeting international regulatory gates.
- **Academia & Workforce:** Need for new skills in circular systems and data analytics. **Key focus:** High-value green tech jobs.
- **Society & Environment:** 62M tons of e-waste (2022). **Key focus:** Intergenerational equity and resource preservation.

1.4 CONCLUSION: A Framework for 21st-Century Leadership

Solving this paradox requires a holistic integration of **Strategic Competencies**:

1. **Ethical Reasoning:** Balancing economic growth with security and environmental justice.
2. **Cross-Sector Collaboration:** Aligning industry, government, and academia.
3. **Agile Strategic Planning:** Adapting to the 2024-2028 regulatory window.
4. **CSR-Integrated Business Models:** Making circularity profitable and scalable
5. **Digital & AI Fluency:** Leveraging blockchain for DPP compliance and AI for yield optimization.
6. **Transparency Governance:** Building verifiable, audit-ready circular systems.

The semiconductor industry cannot meet CHIPS Act goals or remain globally competitive with a linear model. Circular intelligence is no longer optional; it is the strategic prerequisite for technological sovereignty, ethical credibility, and long-term leadership.

The shift is clear:

TAKE → MAKE → WASTE → REPEAT

↓

RECOVER → LEARN → REDESIGN → REGENERATE

This isn't recycling—it's transforming semiconductor manufacturing into an intelligent closed-loop system where every atom and every data point creates lasting value.

SECTION 2 . EVIDENCE-BASED ANALYSIS

The ChipCycle facility operates as a dual-output ecosystem, simultaneously recovering strategic raw materials and generating industrial failure intelligence. The following workflow details the end-to-end transformation of semiconductor scrap, ranging from foundry wafers to consumer electronics, into certified commodities and actionable data.

Stage 1: Inbound Logistics and Digital Initialization

The lifecycle begins immediately upon the arrival of scrap batches at the facility. To ensure strict chain-of-custody, every inbound batch is tagged with active RFID trackers. Simultaneously, the system instantiates a Digital Passport on a private, permissioned blockchain ledger. This records the material's provenance, such as the specific foundry or fab of origin.

Stage 2: Smart Segmentation

The scraps enter the primary conveyor line for automated segmentation. This stage relies on high-resolution optical cameras utilizing Computer Vision (YOLO v12) algorithm to perform real time analysis, distinguishing between loose ICs, whole wafers, or PCBs.

Based on this sensor data, a robotic arm equipped with RGB-D sensors and custom extraction tools executes the disassembly of convenient components. It locates and unscrew or desolder specific components to isolate them from the main PCB. Then, the system automatically splits the material into two distinct processing paths:

- **The Intelligence Line:** High-value targets, such as defective wafers, ASICs, Processing Units, are gently routed to the Forensic Lab for non-destructive testing.
- **The Bulk Line:** Standard consumer PCBs and low-grade scrap are routed directly to the bulk recovery system.

Stage 3: The Intelligence Forensic Lab

This stage represents the core value differentiator of the ChipCycle platform. Instead of immediately destroying high-value scrap, the facility performs an "industrial autopsy" to harvest failure data.

For complex integrated circuits and advanced chips packaging, the system utilizes Ptychographic X-ray Computer Tomography (PXCT) capable of volumetric scanning at the sub 20 nm scale. A 3D Convolutional Neural Network (3D-CNN) processes the data to detect internal structural failures—

such as cracked Through-Silicon Vias (TSVs), solder bridging, or micro-voids—without physically opening the package.

For semiconductor wafers, the process utilizes Atomic Force Microscopy (AFM) to perform nanoscale surface inspection. To make this technical data accessible to clients, a Vision-Language Model (VLM) synthesizes the imagery and sensor readings into a natural language report. This automated report diagnoses the topology to classify specific defects and is sold back to the foundry as R&D intelligence.

Stage 4: Intelligent Material Recovery

Once the Forensic Lab has analyzed the high-value scrap, the bulk stream undergoes physical shredding to a controlled particle size of 10 mm to expose internal surfaces. The shredded output goes through a Dense Medium Separation process where materials are separated by gravity. Light materials such as plastic and fiberglass dust are lifted away for polymer recycling, leaving behind a metal and rare earths concentrate.

This stream is then analyzed by inline X-Ray Fluorescence (XRF) sorters. These read the atomic signature of each shred in milliseconds. Pneumatic actuators accurately target and separate pieces containing copper, aluminum, and precious metals at throughputs reaching 15 tons per hour.

Finally, the system's Intelligent Hybrid Recovery System directs these sorted shreds to a specific decomposition method depending on the material identified. Whether it may be electrodeposition, bioleaching, or soft chemistry.

Stage 5: Certification and Market Re-entry

The workflow concludes by closing the loop on the blockchain ledger. A final Certificate of Recovery (CoR) is created for the recovered materials. This digital token serves as a guarantee of quality and compliance, embedding critical data directly into the transaction:

- **Purity Metrics:** Validated via mass spectrometry.
- **Carbon Footprint:** A calculated metric of CO2 savings compared to virgin mining.
- **Full Provenance:** A complete history linking the refined ingot back to the scrap arrival.

This digital certification allows buyers to prove compliance with regulations like the EU Critical Raw Materials Act, turning the recovered material into a premium, verified "green" commodity.

2.1 PILLAR 1: Intelligent Hybrid Recovery System (AI-driven)

2.1.1 Challenge Statement and Background

Metal recovery has plenty of methods however each process involves different amount of technologies requirements and retrieve a different purity grade such pyrometallurgy, hydrometallurgy or bio-hydrometallurgy (Jadhav & Hocheng, 2012). Performing the choice fully relying on humans lead to slow decision and most likely sub optimal choices.

Artificial Intelligence have been applied on the extraction process to dismantle circuit printed boards, being able to identify components and adjust dynamically within sensors mechanical shredding, pyrolysis and hydrometallurgy (Mohsin et al., 2025) or retrieve favorable conditions for bioleaching (Mokarian et al., 2022). However applying different methods to gather and return the highest amount of source material without harming the environment is a crucial problem to optimize.

Chipcycle proposes an intelligent system capable of recognizing scrap and gather the greatest techniques to treat the industrial wastes, meanwhile the standards and SDGs are still involve. Through the methodology is discussed the set of selection of techniques employed and the artificial intelligence system approach.

2.1.2 Proposed Methodology

Electrodeposition

Electrodeposition technique for gathering materials is a well-known and explored method, their first article regards up to 1905 according to the Scopus Database (Delgado et al., 2021). Along the time its used has been long extended becoming a mature technique developed in plenty of applications such water treatment for water resources, introduced on liquid electrodeposition with organic ionic salt to enhance thermal and chemical stability and bio electrochemical utilizing living systems employing microorganisms or plants to contribute to the oxidation process splitting into fuel cells and electrolysis cells (Delgado et al., 2021)..

Electrodeposition work employing solid material on the cathode of an electrochemical process intervening non external agents besides electrical current (Siwal et al., 2023). Other process for

material extraction relies on electrocoagulation to handle inorganic material pollution, electrodialysis making use of membranes within anode and cathode to target certain material and slurry electrolysis using an electrodeposition (Siwal et al., 2023).

These reactions are made through electrolytic cells, galvanic cells occur spontaneously in presence of an external conductor within the electrode, electronic cells however require external voltage, therefore in the anode occurs the oxidation losing electrons (Brown et al., 2017) thence cathode occurs reduction gaining electrons. In galvanic cells the cathode is positive with the anode, nevertheless over electronic cells the cathode is negative with the respect of the anode (Bard et al., 2022).

Bioleaching

Bioleaching employ microorganism to perform a chemical solve produced by their metabolism or products on their metabolism, the chosen microorganism goes according to conditions such Ph levels or temperature, bacteria and specially fungi is used for their volume of generation reducing cost (Rendón-Castrillón et al., 2023).

The process of bioleaching resides on extracting solute by performing a biosorption and desorption process to finally discompose of the metal. On the biosorption phase, columns of beds are inserted with immobilized microorganisms on matrices producing concentrated organic (HCl, H₂SO₄, HNO₃) or non-organic salt solutions (NaNO₃, Ca(NO₃)₂, sodium citrate) on a process named desorption. Finally, the remaining solution concentrates the metal and forced through chemical precipitation to overcome a solid state (Rendón-Castrillón et al., 2023).

In industrial scale it is utilize bioleaching heap process. Sprinklers distribute a solution with reagents and bacteria that feed from the mineral elementals to separate the metal particles (Saldaña et al., 2023). China built the first heap leaching in 1997 for commercial purposes due the limitations on copper distributions allocated 88% in small scale copper mines (Yin et al., 2018).

Molecular Recognition Technology

Perform metal recognition designing a specific macrocyclic compound targeting specific ions, an macrocyclic compound consist in a large ring-shaped structured able to change size and bind to guest ions due its selective complexation (Zhang et al., 2021; Huang et al., 2024).

The process involves applying chemical ligand bands on solid supports such silicon gel or polymer substrates dismissing non organic solvents and enhancing green chemistry, requiring a loading phase applying the chemical product, applying elution solution and finally washing the remains (Izatt et al., 2016).

Molecular Recognition Technology have shown success on industry scales developing products such SuperLig and applied by international metal companies corresponding Tanaka Precious Materials (TKK), Impala and Koyo (Izatt et al., 2016). The environmental impact results in raffinate streams the table without further recovery process possessing a minimal carbon footprint (Izatt et al., 2016).

AI Driven Recovery System

The proposed system integrates a modular decision intelligence system where supervised models are used to estimate process outcomes and a reinforcement policy prescribes optimal recovery process based on rules and constraints for regulatory compliance.

The decision making process activates the closest experts regarding each field of knowledge being in this case the recovery methods to ensure and select the best parameters for metal extraction, this enhances saving computational resources by only activating the required knowledge required in each case.

Hierarchical Reinforcement Learning

Decides the experts that should attend the following batch of scrap the model employed Deep Q-Network (DQQ), the reinforcement learning model utilizes the following function to maximize.

$$f(x) = (Recovery, Energy, Time, Purity, EnviromentalImpact)$$

$$\alpha, \beta, \gamma, \delta, \varepsilon \in [0, 1], \alpha + \beta + \gamma + \delta + \varepsilon = 1$$

Simplified Notation

$$R = \sum_i w_i x_i$$

Expert Base Models

Supervised models trained in each extraction process, altogether on an ensemble soft voting classifier regarding their confidence value to compute a discrete value representative of the outcome and the total penalties and constraints.

The expert models are integrated within machine and deep learning model such artificial neural networks, random forests and Bayesian networks to estimate risk. The final prediction determinates the sequential structured methods and their parameters, for example temperature, ph for the bioleaching process (Huang et al., 2024).

2.1.3 Expected Results

Improved Metal Recovery Efficiency

The AI-driven decision system is expected to increase overall metal recovery rates by dynamically selecting the most suitable extraction technique for each scrap batch gathering a consistently higher recovery percentages across variable scrap compositions.

- Higher recovery yields compared to single-method or static-process approaches
- Improved capture of critical and precious metals from heterogeneous scrap streams
- Reduced loss of valuable materials due to suboptimal process selection

Reduced Environmental Impact and Regulatory Alignment

The optimization objective explicitly incorporates environmental impact as a weighted decision variable, enabling greener process selection, the system supports compliance with sustainability standards.

- Lower energy consumption compared to high-temperature pyrometallurgical routes
- Reduced use of aggressive chemical reagents
- Minimization of secondary waste streams and toxic effluents

Adaptive and Scalable Recovery Architecture

The hierarchical reinforcement learning framework enables the system to learn continuously from operational data, different experts can be managed and integrated to improve the architecture requiring fine tuning on the reinforcement model to adapt the choice.

2.2 PILLAR 2: Digital Material Passport Ecosystem (blockchain)

2.2.1 Challenge Statement and Background

The "Black Box" of Electronic Waste

Despite being the most technologically advanced industry in the world, the semiconductor sector faces a critical paradox in its waste management. Recent academic literature agrees that the current linear model is unsustainable and opaque.

According to the bibliometric analysis by Naranjo-Paez et al. (2025), research on *e-waste* has grown exponentially in response to an unmanageable volume of waste, yet the infrastructure to manage it remains fragmented. This "Black Box" is defined by three systemic failures:

- Opacity in Reverse Logistics and the Informal Sector: Once materials leave the factory, the chain of custody is broken. Casas-Merchán and Toro-Calderón (2020) warn about the strong involvement of informal organizations in waste management, which prevents guaranteeing ethical standards. Pérez Yusty et al. (2025) reinforce this by pointing out that without

technological traceability, it is impossible to certify whether a material was truly recycled or diverted to illegal landfills, creating a massive reputational risk.

- **Supply Chain Fragility (Scarcity):** The complexity of global supply chains and the "Just-in-Time" model have created extreme vulnerability. Guerrero Sánchez and Pérez Mayo (2022) argue that this complexity causes an artificial scarcity of critical materials; the industry is unable to recover its own valuable resources (such as rare earths) to reintroduce them into the cycle, generating economic losses that Cayumil Montecino and Adasme Valdés (2018) identify as key challenges in *e-waste* processing.
- **Inefficient Data Management:** Currently, validation relies on static documents vulnerable to forgery. The literature converges on the fact that the lack of digital integration is the main barrier to the circular economy (Pérez Yusty et al., 2025).

2.2.2 Proposed Methodology

To solve the trust problem identified in the background, our methodology utilizes Blockchain technology. As defined by Souza (2024), the integration of Blockchain with IoT (Internet of Things) is the technical enabler necessary to ensure data veracity in modern supply chains.

Our methodology relies on a decentralized ledger with three unique characteristics:

1. **Decentralization:** Replicating information across multiple nodes to prevent single points of failure.
2. **Immutability:** Ensuring records are mathematically impossible to edit or delete once registered.
3. **Transparency:** Allowing real-time auditing of the material's history.

Our Solution: ChipCycle Intelligence Platform

We propose the implementation of the **Digital Material Passport**, a solution that transforms physical waste into a traceable digital asset.

Value Proposition & Implementation Strategy:

- **Integrity Verification:** We will eliminate *greenwashing* by assigning unique cryptographic tokens to each batch of recovered material.
- **Circular Integration:** The system tracks the reintroduction of semiconductor-grade materials (99.99% purity) directly back to the production line.

- **Regulatory Automation:** The platform is architected to automatically comply with emerging regulations, specifically the EU Digital Product Passport requirements.

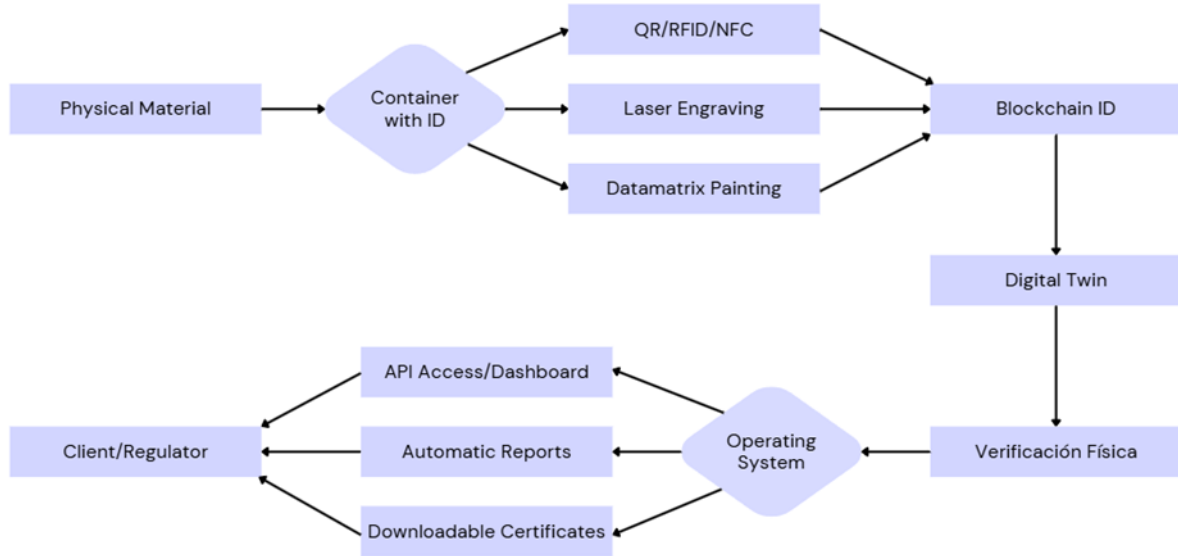


Figure 1 Block Diagram of the Proposed System

2.2.3 Expected Results

Economic Viability and Return Analysis

To validate the sustainability of the model, we have projected the following financial and operational outcomes based on strategic industry reports (Intel, Apple, Bain & Co).

Optimized Cost Structure:

- **Initial Investment (CapEx):** While infrastructure development requires initial capital, our SaaS strategy reduces the hardware burden for the client.
- **Operating Costs (OpEx):** By utilizing *Proof of Stake* consensus protocols, we expect to keep energy consumption and operating costs marginal, ensuring financial and environmental sustainability.

Projected Sources of Value and ROI:

1. **Access to the "Green Premium" Market:** We expect to capture significant additional revenue through price premiums. Reports from **Bain & Company** (The Visionary CEO's Guide to Sustainability, 2023) confirm that corporate buyers are willing to pay a significant

premium for inputs that guarantee *Net Zero* goals. For example, **Apple** already prioritizes "100% recycled and certified gold" (Apple Environmental Report 2023), validating the market demand for ChipCycle's certified materials.

2. **Compliance Cost Efficiency:** We anticipate a reduction in operational costs related to auditing. Manufacturers like **Intel** invest massive resources in auditing suppliers (Intel CSR Report 2024-25). Our platform automates this validation, transforming hours of manual auditing into instant digital verification.
3. **Systemic Value Recovery:** We expect to capture economic value currently lost in the system. The **Global E-waste Monitor** indicates massive annual losses from unrecovered materials; ChipCycle's traceability ensures this "evaporated" value is retained within the supply chain.

Conclusion: The expected result is a financially robust model that transforms regulatory compliance from a "cost center" into a revenue-generating competitive advantage.

2.3 PILLAR 3: Prescriptive Industrial Intelligence Platform

2.3.1 Challenge Statement and Background

The semiconductor industry is currently struggling to adequately characterize and diagnose complex structures uninvasively. Modern nanoelectronics have advanced to a point at which it is impossible to image entire devices and their interconnections non-destructively because of their small feature sizes and the complexity of three-dimensional structures (Holler et al., 2017).

Hidden Complex Structures: Modern 3D packaging technologies, such as Through-Silicon Vias (TSVs) and stacked dies, create internal structures that are inaccessible to surface inspections and standard optical microscopy. Voids and disconnects buried deep within these stacks lead to "weak opens" that are difficult to detect without destroying the package (Wang & Wang, 2014).

Data Loss in Scrap: The proliferation of electronic devices has accelerated e-waste generation, estimated at over 54 million tons in 2019 (Rahimi et al., 2026). Currently, foundries treat defective wafers as material waste rather than information assets. There is no efficient, automated workflow to extract forensic manufacturing data from this scrap to identify upstream root causes of failure (e.g., lithography drift or deposition errors) before the materials are recycled (Rundo et al., 2024).

Limitations of Current Defect Detection

Current industrial quality control relies heavily on Electrical Wafer Sorting (EWS), which generates binary defect maps (working vs. non-working dies). While these maps reveal patterns like "Center," "Donut," or "Edge-Ring" defects, interpreting the root cause remains a complex hypothesis-driven task often requiring human intervention (Rundo et al., 2024).

2.3.2 Proposed Methodology

ChipCycle's compromise is to bridge this gap by implementing a non-destructive, high-throughput forensic intelligence layer that converts "scrap" into actionable manufacturing insights before material recovery. Our Forensic Lab depends on the integration of a dual cutting-edge imaging technique plus machine learning algorithms:

- **X-Ray Ptychography (PXCT) + 3D Convolutional Neural Network:**

Unlike standard X-ray imaging, which is limited by lens resolution, ptychography is a coherent diffractive imaging technique. It can resolve 3D Integrated Circuit structures down to 14.6 nm non-destructively (Holler et al., 2017). This allows for the visualization of internal transistor gates, interconnects, and manufacturing faults in 3D without delayering the chip.

Convolutional Neural Networks (CNNs) have proven highly effective in automating defect recognition. For example, deep learning models can classify wafer defect map patterns with over 95% accuracy and detect voids in TSVs from 2D X-ray images by analyzing block features (Wang & Wang, 2014). We propose a novel approach of combining the imagery of X-Ray Ptychography and the power of 3D-CNNs to automatically and time-cost effectively detect defects on Integrated Circuit Packages.

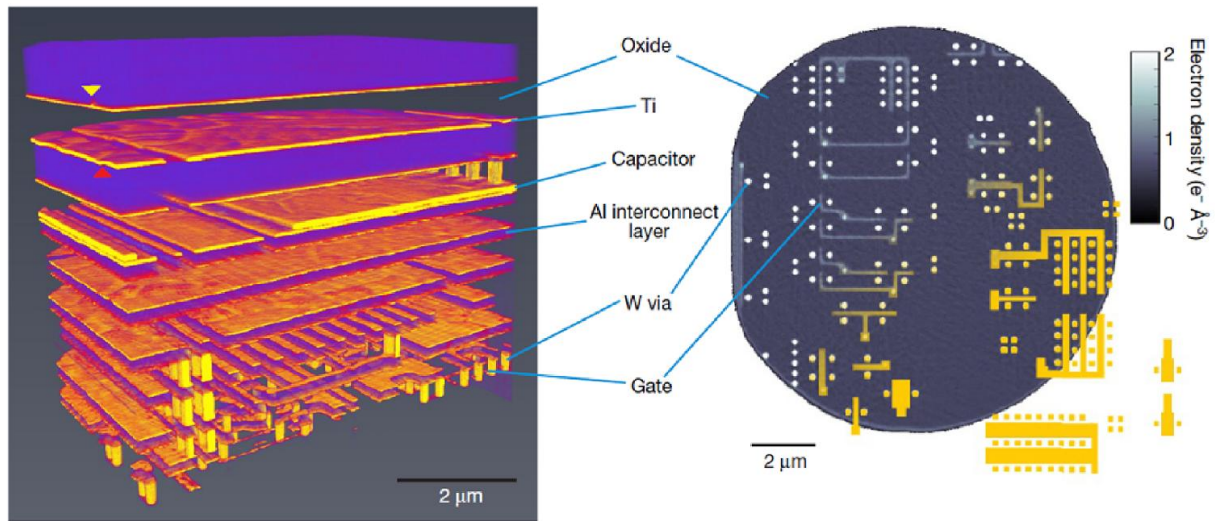


Figura 2 Nano-resolution X-ray imaging with ptychography for ICs (Wang & Wang, 2014).

- **Atomic Force Microscopy (AFM) + LLM-Driven Decision Support:**

Atomic Force Microscopy is an excellent instrument for mechanical characterization and high-resolution surface topography mapping. AFM can be employed to probe and investigate silicon wafer surfaces for defect or contamination detection. (Biswas et al., 2025) The complexity of nanoscale defects, such as those found via Atomic Force Microscopy (AFM), often requires specialized expertise. Large Language Models (LLMs) have emerged as powerful assistants capable of interpreting defect data and providing real-time, context-aware corrective guidance to technicians, bridging the knowledge gap in failure analysis (Biswas et al., 2025).

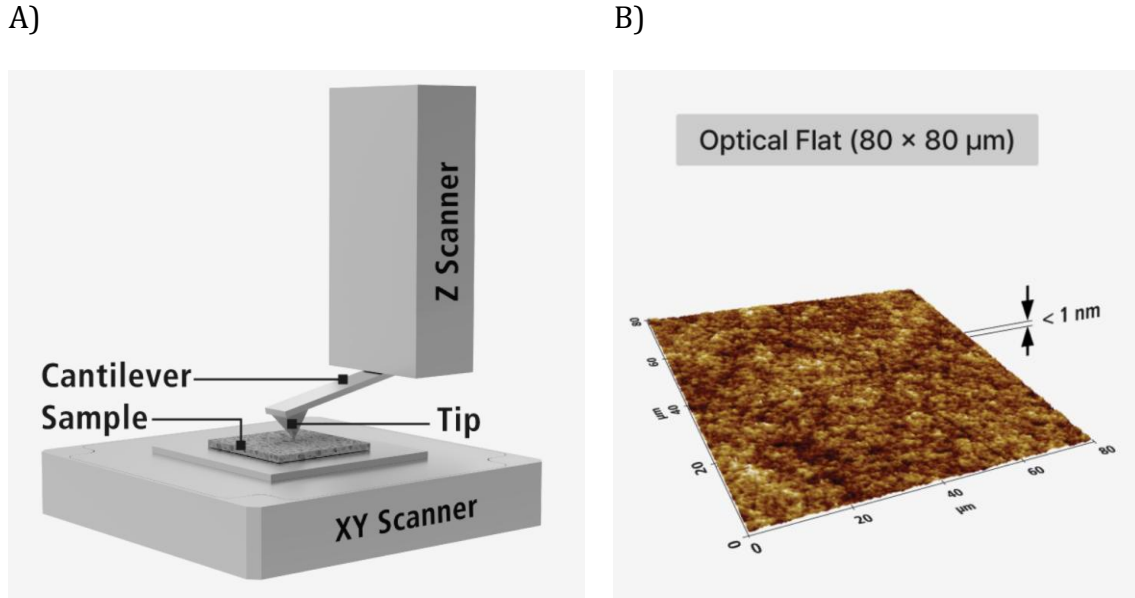


Figure 3 - A) Atomic Force Microscopy working principle illustrated. A <5nm apex radius tip scans the sample while the cantilever records topological information from deviation of minuscule Van der Waals forces. B) Topological information achieving picometer resolutions obtained by Atomic Force Microscopy (Biswas et al., 2025).

By integrating these technologies, ChipCycle moves beyond simple "recycling" to establish a forensic feedback loop, creating value from the information stored within the defect before reclaiming the material itself.

2.3.3 Expected Results

Accelerated Yield Ramping and Process Optimization:

The implementation of ChipCycle's Intelligence Platform is expected to fundamentally alter the semiconductor quality control landscape by converting "scrap" into actionable forensic data. By providing high-volume, statistical failure analysis data from the scrap stream—data that is currently lost to mechanical shredding—we enable fabs to identify subtle, systemic process drifts that their internal sampling might miss.

This feedback loop allows engineering teams to correlate specific defect signatures (e.g., micro-voids in TSVs or specific particle contaminants) with batch origins, facilitating faster root-cause corrective

actions. We project that integrating this external forensic loop could reduce yield ramp-up periods for new nodes by weeks, translating to millions of dollars in operational savings and recovered revenue.

SECTION 3 . STAKEHOLDER MAP + CSR PROPOSAL

Stakeholder	Category	Interest	Influence	Impact
Semiconductors Factories	Industrial / Client	Reliable Source Material, Loss Reduction	Very High	Direct
Electronics Manufacturing Services	Industrial / Client	Transparency, Reputation	Medium - High	Indirect
Government	Public / Law Enforcement	National Security, Regulations	Very High	Direct
Investors	Financial	Long Term, Measurable Impact	Very High	Direct
Scrap Suppliers	Industrial	Risk Transfer, Liability Management	High	Direct
Recyclers & Recovery Operations	Industrial	Access Advance Recovery Technology	Medium	Indirect
Environmental NGOs	Social	Transparency, Waste Reduction	Low	Indirect

Table 2 Stakeholders Map

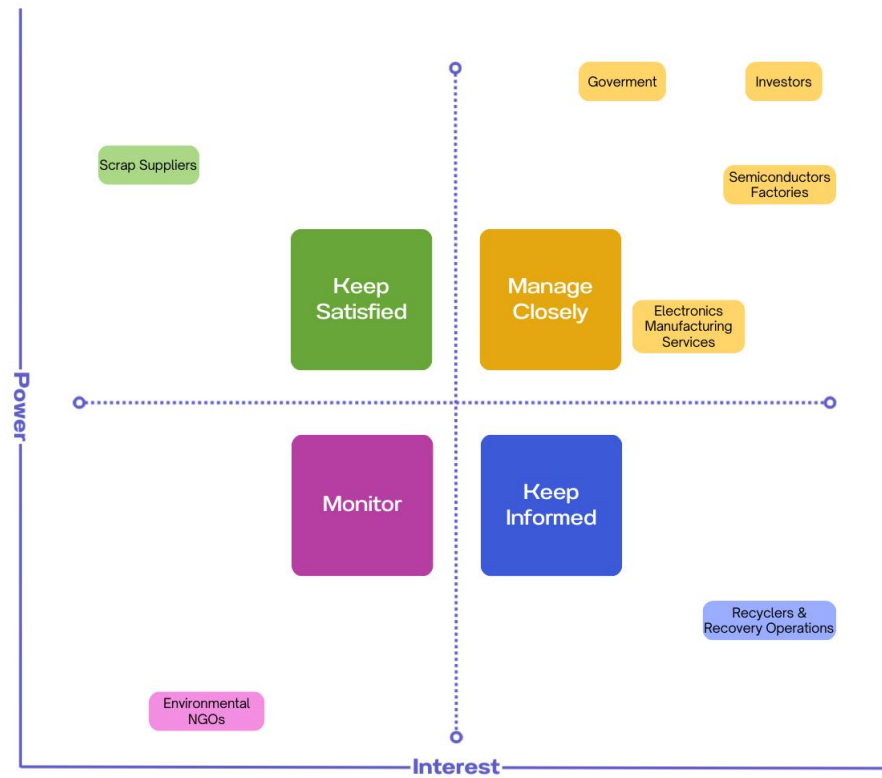


Figura 4 Stakeholders Map

Manage Closely

Investors: Access to financial reports, source management and long-term measurable impact, promote quarter informs displaying the current track goals, advances on portfolio, programs or projects, future predictions and opportunity fields.

Government: Fulfill regulations such U.S. CHIPS Acts, EU Product Passport, California SB-253, inform about long term contracts and operations.

Semiconductors Factories: Compliance delivery schedules and demand, stay up to date regarding their new requirements and projections.

Electronics Manufacturing Services: Provide clear information about regulations compliance, promote communication and notice their new market tendencies.

Keep Satisfied

Scrap Suppliers: Generate a trustworthy partnership, become their first option to process their waste, and deliver risk.

Keep Informed

Recyclers and Recovery Operations: Share resume of recovery process, closely up to date progress, encourage information exchange.

Monitor

Environmental NGOs: Visualize feedback and observe trends.

3.1 CRS Proposal

ChipCycle is compromised within their planet, people and society, the four pillars of standard Corporate Social Responsibility are strongly applied in the entire organization, sustainability and correct use of emergent technologies to treat and recover materials.

Pillar 1: Environmental Stewardship & Circular Economy

Initiatives:

- Recovery of critical and precious metals from post-industrial and end-of-life semiconductor scrap.
- Reduction of CO₂ emissions and toxic tailings compared to primary mining.
- Implementation of intelligent recovery methods optimized for energy efficiency and environmental impact.

Impact:

- Lower Scope 3 emissions for partner fabs and OEMs.

- Reduced contribution to global e-waste streams.
- Preservation of critical raw materials.

Pillar 2: Transparency, Traceability & Responsible Governance

Initiatives:

- Deployment of a Digital Material Passport for every scrap batch and recovered material.
- Blockchain-based chain-of-custody records ensuring immutability and auditability.
- Issuance of a Certificate of Recovery (CoR) containing purity metrics, provenance, and carbon savings.

Impact:

- Elimination of greenwashing risks.
- Automated compliance with EU Digital Product Passport and ESG reporting requirements.
- Increased trust across the supply chain.

Pillar 3: Secure Handling & Ethical Data Management

Initiatives:

- Controlled processing of defective wafers and ICs to prevent intellectual property leakage.
- Secure forensic analysis prior to material recovery.
- Federated and permissioned data architectures ensuring client IP never leaves authorized environments.

Impact:

- Protection of proprietary semiconductor designs.
- Increased willingness of fabs to participate in circular recovery programs.
- Alignment with Responsible AI and data governance principles.

Pillar 4: Social Impact & Workforce Development

Initiatives:

- Creation of high-skilled green technology jobs in AI, robotics, and material science.
- Collaboration with universities and research centers for training and applied research.
- Formal integration of recycling partners into compliant, safe, and ethical operations.

Impact:

- Workforce upskilling for the circular semiconductor economy.
- Reduction of informal and unsafe recycling practices.
- Strengthened local innovation ecosystems.

SECTION 4 . AGILE PLAN / TIMELINE

To execute the multi-disciplinary roadmap of the ChipCycle Intelligence Platform, the leadership team is structured around four critical domains: **Hardware Infrastructure, Software Architecture, Artificial Intelligence, and Strategic Operations.**

Edman Mendoza: Hardware Engineering Lead (CTO)

Responsible for the selection, installation, and synchronization of the physical sensory and sorting machinery.

- **Key Responsibilities:**
 - **Sensor Selection:** Lead technical procurement for the **Inline XRF** and **Micro-CT** system, ensuring resolution meets the "10nm defect detection" threshold.
 - **Robotics Integration:** Design the kinematic control loops for the Robot array, and synchronization with the conveyor belt speed (<10ms latency).
 - **Safety & Compliance:** Oversee radiation safety protocols for X-Ray equipment and worker safety.

Alejandro Diaz: Software Architecture Lead (CIO)

Responsible for connecting physical sensors to the cloud and ensuring data immutability.

- **Key Responsibilities:**
 - **Digital Passport:** Architecting the **Private Blockchain (Hyperledger)** infrastructure to manage the "Certificate of Recovery".
 - **Sensor Fusion Pipeline:** Building the low-latency API that ingests raw data from Edman's sensors and feeds it to Daniel's AI models.
 - **Cybersecurity:** Implementing the **Federated Learning** protocols to ensure client (Foundry) IP never leaves their local servers.

Daniel Guzman: AI & Computer Vision Lead (CDO)

Responsible for training the models that turn raw pixels into valuable failure intelligence.

- **Key Responsibilities:**
 - **Vision Systems:** Implementing the **YOLO v12** architecture for high-speed scrap classification and turning **XRF** scans into material recognition.
 - **Forensic Intelligence:** Developing the **3D-CNN** for Micro-CT volumetric analysis and the **Vision-Language Model (VLM)** for automated reporting.
 - **Data Strategy:** Managing the **Synthetic Data Generation (GANs)** pipeline to overcome the lack of initial training data for rare defects.

Fabiola Medina: Business & Operations Lead (CEO/CFO)

Responsible for ensuring the technology translates into a profitable, compliant business model.

- **Key Responsibilities:**
 - **Strategic Sourcing:** Securing **Letters of Intent (LOI)** with pilot Foundries and negotiating "Take-Back" agreements for the processed scrap.
 - **Regulatory Compliance:** ensuring the process meets the **EU Critical Raw Materials Act** and obtaining necessary ISO certifications for the "Digital Passport."
 - **Unit Economics:** Managing the budget and defining the pricing model for the "Intelligence Data" service.

4.1 The 'Macro' View: 18-Month Development Roadmap

Our development strategy is divided into four distinct **Program Increments (PIs)**. Each PI consists of twelve 2-week sprints. This allows us to align long-lead hardware procurement (e.g., XRF sensors, Robot Arms) with rapid software iteration.

Target Launch: Q3 2027 (Pilot Deployment with Partner Fabs)

Phase / Dates	Focus Theme	Key Objectives
Phase 1: Foundation <i>(Q1 - Q2 2026)</i>	Architecture & Procurement	• [Hdw] Finalize specs for XRF and Micro-CT; Place orders (6-month lead time). • [AI] Train YOLO v12 on open-source datasets. • [Soft] MVP of Blockchain Ledger (Testnet).
Phase 2: The "Alpha" <i>(Q3 - Q4 2026)</i>	Module Prototyping	• [Integ] "Frankenstein" Prototype: Robot arm successfully guided by Vision System in lab. • [AI] "Synthetic Data" pipeline established to cover missing hardware data. • [Biz] Letter of Intent (LOI) signed with first pilot Foundry.
Phase 3: The "Beta" <i>(Q1 - Q2 2027)</i>	System Integration	• [Hdw] Install Inline XRF & Conveyors; Connect all stations (Sensing →Sorting →shredding). • [Soft] End-to-End Digital Passport trial (Physical scrap → Digital Token). • [Test] 24-hour continuous run test (Stress Testing).
Phase 4: The "Pilot" <i>(Q3 - Q4 2027)</i>	Fab Validation	• [Live] Receive first truckload of real scrap from Partner Fab. • [AI] First "Intelligence Report" sold/delivered to client. • [Cert] ISO & Environmental compliance audits initiated.

Table 3 Timeline

4.2 The 'Micro' View: The Sprint Lifecycle

While the roadmap guides the year, the team operates in **2-week Sprints**. This cadence balances the need for deep work (coding/engineering) with the need for frequent synchronization across the hardware-software divide.

Sprint Mechanics:

- **Duration:** 10 Working Days (2 Weeks).
- **System of Record:** Notion (Kanban Board) & GitHub (Code Repositories).
- **Sprint Goal:** Every sprint must yield a *Potentially Shippable Increment (PSI)*, whether that is a physical prototype, a trained model, or a validated budget.

Below is a snapshot of Sprint 1 to demonstrate our operational cadence during the initial Architecture and Procurement Stage:

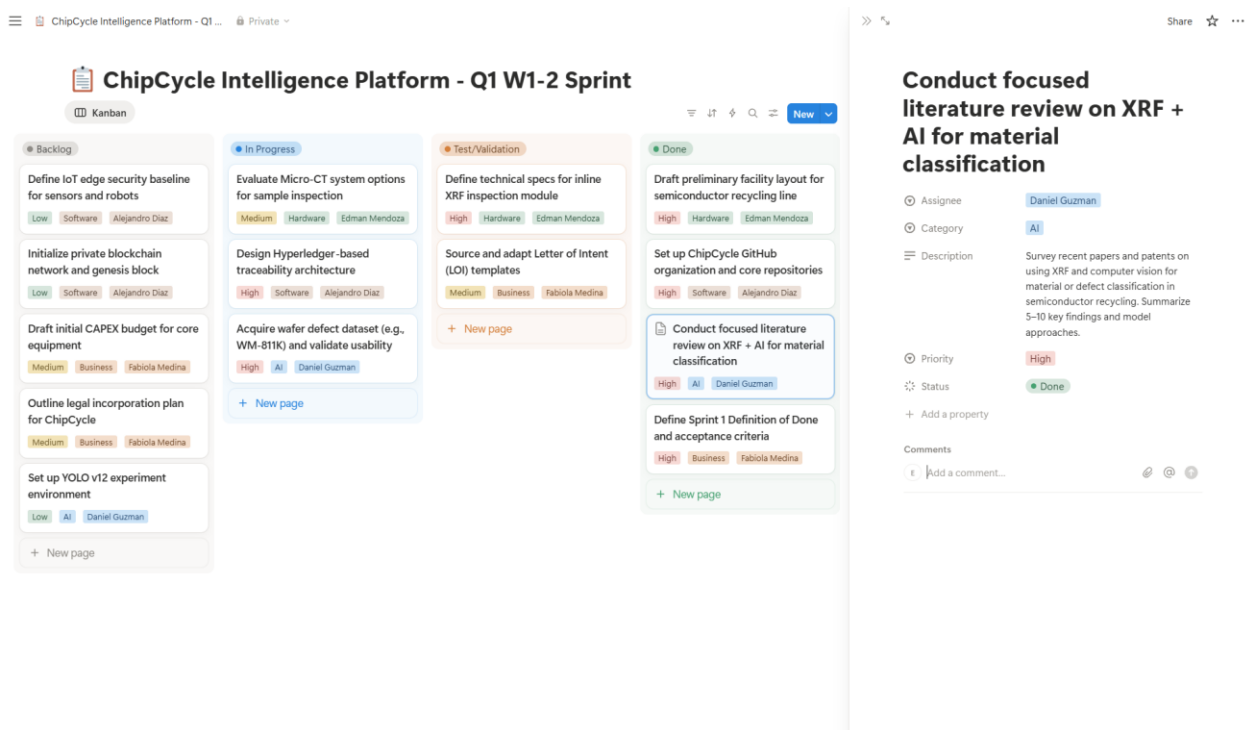


Figure 5 Kanban Board with activities from Q1 Week 1-2 (Sprint 1) made using Notion.

During the 10 working days sprint, the team will be conducting some key Agile ceremonies that ensure cross-department communication, promote process improvement and conflict solving.

Key Agile Ceremonies:

- **Sprint Planning (Day 1 – Monday 09:00 AM):** Defining the sprint goal, clearing priorities, and assigning tasks from the Backlog.
- **The ‘Cross Domain’ Daily Standup (Daily – 09:45 AM):** A 15-minute sync meeting to identify blockers and discuss daily progress.
- **Sprint Review and Retrospective (Day 10 – Friday 14:00 PM):** Inspect the outcome, demonstrate results, and process improvements.

4.3 Change Management: The Decision Record Protocol

In Agile deep-tech integration, adapting to technical reality is inevitable. When a critical impediment arises—such as a hardware-software integration failure or a supply chain delay—the team immediately triggers a "Swarm" session. Instead of waiting for the next scheduled meeting, the relevant domain leads suspend non-critical tasks focusing exclusively on the bottleneck.

To ensure that these deviations are strategic rather than reactive, the team utilizes a formal Architecture Decision Record system. This "Pivot Log" serves as the single source of truth for why the product roadmap evolved over time.

This living document captures the context of the block, the options considered, and the rationale behind the final decision. By treating decision-making as a formal deliverable, the team creates a permanent "institutional memory" that justifies changes to the roadmap, prevents the re-litigation of solved issues, and provides stakeholders with a transparent audit trail of how the product has evolved to meet technical realities.

SECTION 5 . RESPONSIBLE AI & DIGITAL TOOLS NOTE

5.1 Tools Used and Strategic Purpose

In the development of ChipCycle Intelligence Platform, we integrated various AI and digital tools to enhance our research capabilities, optimize workflows, and validate technical concepts.

Tool	Specific Application in Project	Verification Protocol
Elicit	Scientific Literature Review: Used to identify high-impact papers (2020-2025) validating the feasibility of bioleaching and blockchain in circular supply chains.	Cross-Reference: Every key insight extracted by Elicit was manually verified against the original PDF of the paper to ensure context accuracy.
Gemini / ChatGPT	Brainstorming & Drafting: Used to generate initial ideas for business model and to structure the technical arguments for the proposal.	Expert Validation: Generated concepts were reviewed against industry standards (e.g., EU Digital Product Passport regulations) to ensure feasibility.
Canva	Visual Communication: Used to design the stakeholder map and infographic, ensuring complex technical data was accessible to non-expert audiences.	Team Review: All visuals were approved by the team to ensure they accurately represented the underlying data.
Google Scholar	Citation Verification: Used to check the impact factor of journals suggested by Elicit and to format references correctly (APA/IEEE).	Source Check: Verified that cited authors were affiliated with reputable institutions.
Kanban	Agile Project Management: Used to organize the project timeline (Sprints), assign specific roles to members, and track the completion of deliverables.	Synchronous Validation: Card status was only moved to "Done" during live team meetings after a consensus review of the completed task.

Table 2 AI & Digital Tools used

5.2 Information Verification Process: The "Trust but Verify" Protocol

To mitigate the risk of AI "hallucinations" (invented data), we implemented a strict **Three-Layer Verification Protocol**:

- **Source Triangulation:** No critical data point (e.g., "China controls 92% of rare earths") was accepted from a single AI source. We required confirmation from at least one primary source (e.g., USGS, IEA, or a peer-reviewed paper).
- **Sensitivity Analysis:** For our economic projections (ROI), we tested scenarios with variable ranges (conservative vs. optimistic) rather than relying on a single AI-generated number.

- **Human Oversight:** All final text and strategic decisions were reviewed by the team. AI was used as a *co-pilot* for drafting, but the *captain* (final editor) was always a human team member.

5.3 Ethical AI Use Statement

"In developing ChipCycle, we leveraged Artificial Intelligence as a tool to amplify our analytical capacity, not to replace critical judgment. We transparently disclose the use of these tools and accept full responsibility for the accuracy and ethical implications of the final proposal. We have proactively filtered out potential biases in AI-generated data by diversifying our sources across multiple regions (Asia, EU, Americas)."

SECTION 6 . Teamwork Evidence & Collaborative Methodology

6.1 Ideation and Agreement

In Our collaboration began with an in-person meeting on **Monday**, where we discussed potential project topics that aligned with the course objectives.

Fabiola suggested the concept of an "Intelligent Recycling Platform." She highlighted the opportunity to combine three key areas: recovering expensive materials, ensuring traceability through digital passports, and selling "scrap intelligence" data to foundries.

The team agreed to pursue this idea because it addressed a real-world problem. We recognized that the industry is currently losing money on waste and facing pressure from new sustainability regulations, making this a relevant and timely solution.

6.2 Role Assignment

Once the topic was defined, we assigned roles based on each member's personal strengths and interests.

Technical Development: We divided the solution's three main features among us:

- **Daniel** took charge of the **Material Recovery**, researching smart algorithms to select the best salvage methods.
- **Alejandro** focused on **Traceability**, designing the Blockchain architecture for the digital passport.
- **Edman** handled the **Intelligence Platform**, defining how to extract data from scrap using sensors like X-Ray CT scans.

Strategic & Operational Tasks:

- **Fabiola** wrote the **Challenge Statement** and background, drawing on her initial research into the regulatory landscape. She also produced the **Video Pitch**.
- **Edman** developed the **Agile Timeline** and project plan, applying his previous experience with startup MVPs to structure the team's workflow.
- **Alejandro** created the **Responsible AI Guidelines**, advising the team on how to use AI tools ethically and correctly cite sources.
- **Daniel** refined the **Stakeholder Map**, ensuring that Corporate Social Responsibility (CSR) aspects were fully covered.

6.3 Communication and Tools

To keep the work synchronized, we used a mix of in-person and digital channels.

- **Tools:** We collaborated on a shared **Microsoft Word** document via OneDrive and used **Canva** for the slides. A **WhatsApp** group kept everyone connected for quick updates and questions throughout the week.
- **Mid-Week Check-in:** We met again online on **Wednesday evening** to share our individual progress, troubleshoot any issues, and merge our separate sections into a cohesive project.

6.4 Presentation Strategy

For the final pitch, we assigned roles to match our specific skills:

- **Speakers (Alejandro & Daniel):** Chosen to present the main speech due to their strong stage presence.
- **Q&A Lead (Edman):** Chosen to answer questions, as his work on the technical roadmap gave him a deep understanding of the project's feasibility.
- **Production Lead (Fabiola):** Chosen to manage the slide design and visual content, leveraging her knowledge of the compliance regulations to ensure the visuals were accurate.

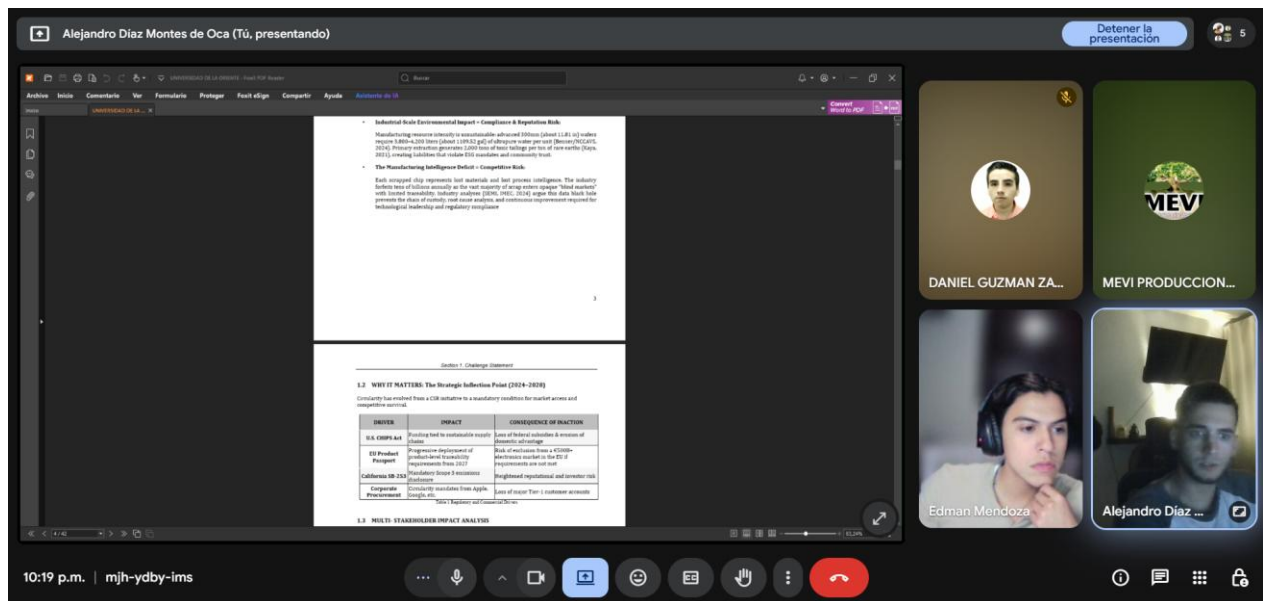


Figura 5 Final Meeting

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SECTION 8 . Appendices: Individual Reflections

8.1 Fabiola Medina Villaseñor – Individual Reflection

My global management experience was profoundly enriched through this diploma's emphasis on Ethics, English, and Employability. Collaborating with talented, younger classmates from diverse backgrounds reinforced my belief in multidisciplinary teams as engines of innovation.

The intensive one-week challenge developing ChipCycle demanded rapid integration of ethics, sustainability, and technical viability. We prioritized environmental justice and transparency alongside profit, honing my agile leadership under pressure as we balanced goals with practical execution.

This experience directly strengthens my employability for the green transition. Grounded in the program's core pillars of **Ethics, English, and Employability**, I gained critical tools to translate semiconductor challenges into actionable, innovative business strategies. The ethical framework compelled me to design for true sustainability at the core of operations, not as an afterthought. I'm grateful to my team and instructors for a rigorous environment that reaffirmed a vital lesson: the most viable solutions are built on ethical foundations and collaborative intelligence.

8.2 Edman Gamaliel Mendoza Leyva – Individual Reflection

Dedicating my time to this course has helped me identify the critical soft skills required for a fast-paced, high-tech global sector like the semiconductor industry. By prioritizing my communication, problem-solving, and workload management skills, I feel I have achieved a level of preparation for the real world that complements my technical background in ways I hadn't before.

This journey improved not only my interpersonal skills but also my "big picture" understanding of the industry's ethical intricacies. I have gained a deeper appreciation for the impact of Corporate Social Responsibility (CSR) decisions and the necessity of complying with international regulations in such a globally impactful field.

Working on the final project allowed me to dive deep into the complex challenges the semiconductor industry faces today. Before this, I hadn't fully recognized the magnitude of the current paradigm shifts and the regulatory pressures weighing on the sector. It was a fulfilling challenge to put together a well-structured, robust solution for managing e-waste by implementing intelligent platforms that respond to actual industry necessities. All within the span of a single week.

Furthermore, working alongside my teammates significantly enriched my collaborative abilities. Learning to recognize individual strengths, receiving and providing assistance, and effectively delegating responsibilities are skills that will serve as a foundation for team projects throughout the rest of my career.

All in all, I am deeply grateful for the experience I have acquired during this diploma. I want to thank the wonderful people I met, not only the professors who delivered valuable insights, but also my colleagues who shared this three-month learning journey with me.

8.3 Daniel Guzman Zavala – Individual Reflection

Through the journey I was able to acquire new knowledge regarding regulations companies must follow given by governments or social responsibility, especially on crucial decision making high members need to decide and sacrifice some scenarios proposed on unequal conditions, strictly deadlines and unpredictable incidents.

Along the course the most interesting topics were regarding conflict management and therefore negotiation steps since they are applied on every human daily to achieve their own goals and how it needs to exist an equal power to do not over concede on the long term, nevertheless I found more interesting meeting and hearing people experiences within the units and how their own experiences resemble the acquired knowledge for future intervenes, although being something unlikely to be match or quantitative, this type of activities promote people with quite similar background to group and go further.

Concerning the final project, I found enjoyable and always will be to engage on technical aspects to provide and discuss with teams the different approach at the same problem, it was a pleasure to learn about chemical process to extract minerals and complement it through my acquired knowledge on the artificial intelligence field and skills to communicate results.

Pleasure to stand and work with reliable and proficient people.

8.4 Alejandro Diaz Montes de Oca – Individual Reflection

Participating in the development of the ChipCycle Intelligence Platform has been a pivotal experience for my career. As an engineer accustomed to technical problem-solving, this project challenged me to think strategically about business ecosystems. I enhanced my employability by bridging the gap between "hard tech" (Blockchain, IoT) and "business value" (ROI, Regulatory Compliance). I learned that in the modern semiconductor industry, technical skills are not enough; the ability to communicate how a technology solves economic and ethical problems is what truly defines a leader.

My research into Pillar 2 (Digital Material Passport) fundamentally changed my understanding of the supply chain. I learned that trust is a technological asset. Discovering how Blockchain can be used not just for finance, but to guarantee ethical sourcing and environmental compliance, was eye-opening. This diploma has equipped me with a "Sustainability-First" mindset, teaching me that the most viable engineering solutions are those that align profit with the well-being of the planet and society.